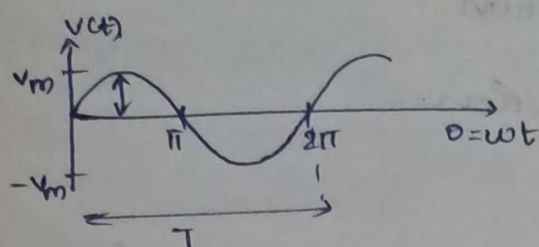


Q6/9

SINUSOIDS & PHASORS

amplitude, time period.

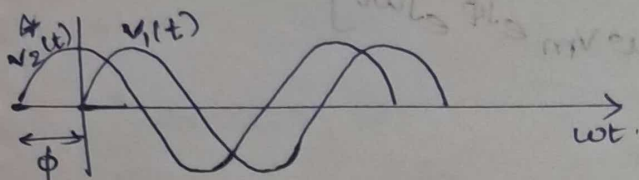
$$v(t) = V_m \sin \omega t$$

$$T = \frac{2\pi}{\omega}$$

* periodic functions—

$$v(t+T) = V_m \sin \omega(t+T) = V_m \sin(\omega t + 2\pi) = V_m \sin \omega t = v(t)$$

$$\Rightarrow v(t+T) = v(t)$$

 $v_2(t)$ is leading $v_1(t)$ by ϕ

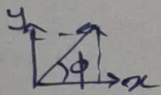
or

 $v_1(t)$ lags $v_2(t)$ by ϕ .

* phasor is applicable only for steady state responses.

only for circuits which work at the same frequency.

$$\bar{Z} = R + jX = R + j\omega L \quad \begin{matrix} \text{resistor} \\ \text{inductor} \end{matrix}$$



$$= R(\cos \phi + j \sin \phi) = R e^{j\phi} \text{ (exponential)}$$

(trig)

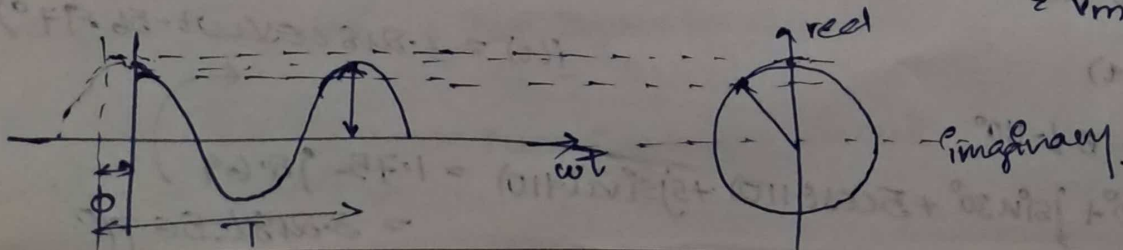
$$v(t) = V_m \cos(\omega t + \phi)$$

$$= \operatorname{Re} [V_m \cos(\omega t + \phi) + j V_m \sin(\omega t + \phi)]$$

$$= \operatorname{Re} [V_m e^{j(\omega t + \phi)}]$$

$$= \operatorname{Re} [V_m e^{j\omega t} e^{j\phi}] = \operatorname{Re} [\bar{V} e^{j\omega t}] \quad \left(\bar{V} = V_m e^{j\phi} \right)$$

$$= V_m \angle \phi$$

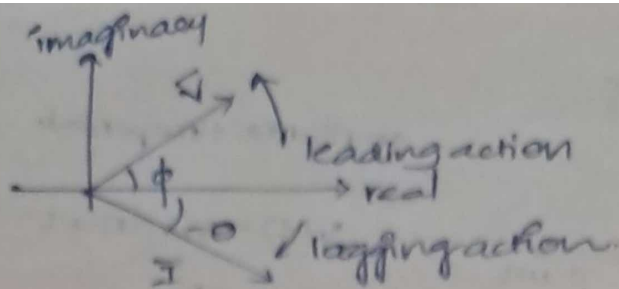


$$v(t) = V_m \cos(\omega t + \phi)$$

$$\bar{V} = V_m \angle \phi$$

$$i(t) = I_m \cos(\omega t - \theta)$$

$$\bar{I} = I_m \angle -\theta$$



→ Always convert to cos form as we're considering it as the real part of a complex number.

$$\frac{dv}{dt} = -\omega V_m \sin(\omega t + \phi)$$

$$= \omega V_m \cos(\omega t + \phi + 90^\circ)$$

$$= \text{Re}[\omega V_m e^{j\omega t} e^{j(\phi + 90^\circ)}] = \text{Re}[j\omega V_m e^{j\phi} e^{j\omega t}]$$

$$\Rightarrow \boxed{\frac{dv}{dt} = j\omega V_m \angle \phi}$$

→ $v(t)$ is time dependent & real.

$\bar{V}$ is not a complex number.

Q Find phasor notation

$$i(t) = 6 \cos(5\omega t - 40^\circ); \quad v(t) = -4 \sin(30t + 50^\circ)$$

$$\text{Sol } i(t) = \text{Re}[6e^{j5\omega t} e^{-j40^\circ}]$$

$$\Rightarrow 6 \angle -40^\circ$$

$$v(t) = -4 \sin(30t + 50^\circ)$$

$$= +4 \cos(30t + 50^\circ + 90^\circ)$$

$$= +4 \cos(30t + 140^\circ)$$

$$\Rightarrow \underline{4 \angle 140^\circ} \Rightarrow +4 \angle 140^\circ$$

$$Q \quad \bar{i}_1(t) = 4 \cos(\omega t + 30^\circ) = 4 \angle 30^\circ$$

$$\bar{i}_2(t) = 5 \sin(\omega t - 20^\circ) = 5 \angle -110^\circ$$

$$\bar{i}(t) = \bar{i}_1(t) + \bar{i}_2(t)$$

$$= 4 \angle 30^\circ + 5 \angle -110^\circ$$

$$= 4 \cos 30^\circ + j \sin 30^\circ + 5 \cos(-110^\circ) + 5j \sin(-110^\circ) = 1.75 - j2.69$$

$$= 3.218 \angle -56.97^\circ$$

$$i(t) = 3.218 \cos(\omega t - 56.97^\circ)$$

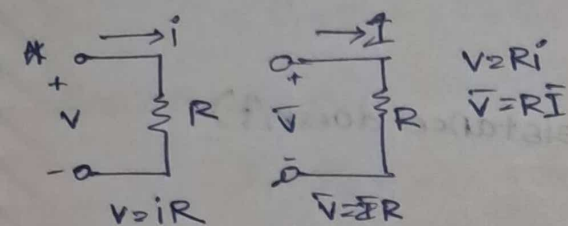
$$\oint 4i + 8 \int i dt - 3 \frac{di}{dt} = 50 \cos(\omega t + 75^\circ)$$

$$4\bar{I} + \frac{8\bar{I}}{j\omega} - 3j\omega\bar{I} = 50 \angle 75^\circ$$

$$\Rightarrow 4\bar{I} - 4j\bar{I} - 6j\bar{I} = 50 \angle 75^\circ \Rightarrow \bar{I} = \frac{50 \angle 75^\circ}{4 - 10j}$$

03/10/23

Phasors and circuit elements:



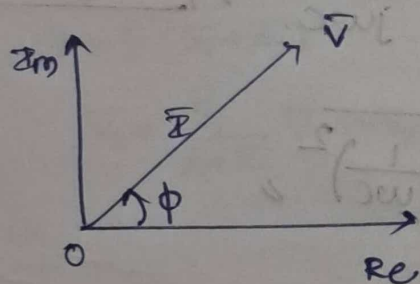
$$i = I_m \cos(\omega t + \phi)$$

$$V = R I_m \cos(\omega t + \phi)$$

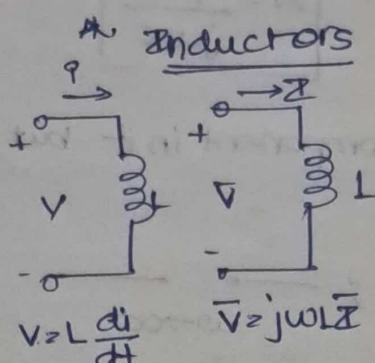
$$V_m$$

$$\bar{V} = V_m \angle \phi$$

$$\bar{I} = I_m \angle \phi$$



(Resistor circuit),



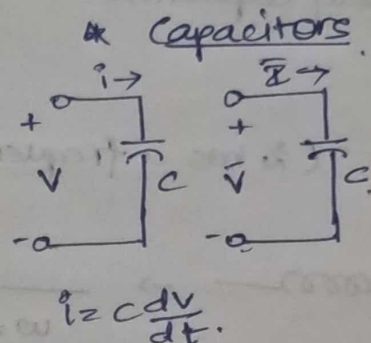
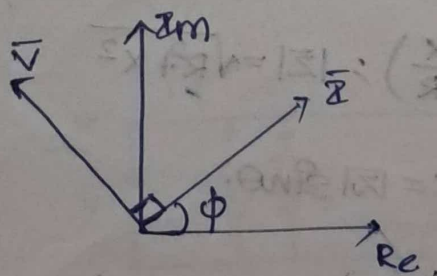
$$V = L \frac{di}{dt}$$

$$= -L\omega \sin(\omega t + \phi)$$

$$= L\omega I_m \cos(\omega t + \phi + 90^\circ)$$

$$\bar{V} = jL\omega I_m \angle \phi$$

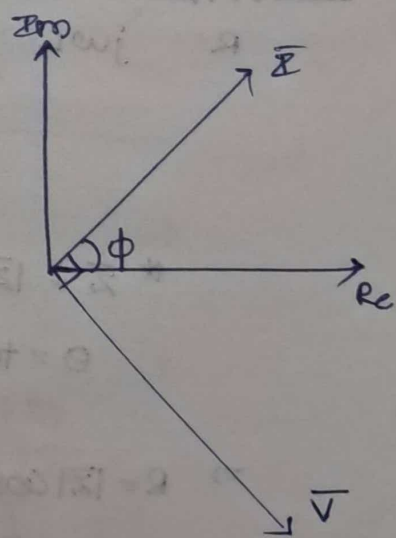
$$\boxed{\bar{V} = jL\omega \bar{I}}$$



$$\bar{I} = j\omega C \bar{V}$$

(OR)

$$\bar{V} = \frac{\bar{I}}{C} \cdot \frac{1}{j\omega} = \frac{-j\bar{I}}{\omega C}$$



$V = RI$ $\bar{V} = R\bar{I} \rightarrow$ Pure Resistor circuit

$V = L \frac{di}{dt}$ $\bar{V} = jL\omega \bar{I} \rightarrow$ Pure Inductor circuit

$V = \frac{1}{C} \int i dt$ $\bar{V} = \frac{\bar{I}}{j\omega C} \rightarrow$ Pure Capacitive circuit.

* Impedance!

$$\boxed{\bar{Z} = \frac{\bar{V}}{\bar{I}}} \rightarrow \text{Basic definition.}$$

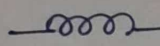
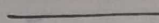
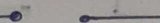
* In a purely resistive circuit,

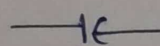
$$\boxed{\bar{Z} = R}$$

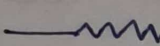
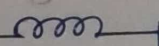
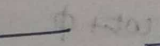
* Purely inductive circuit $\Rightarrow \boxed{\bar{Z} = j\omega L}$

* Purely capacitive circuit $\Rightarrow \boxed{\bar{Z} = \frac{1}{j\omega C}}$

($\bar{Z}$ has a frequency component in it but resistance doesn't)

*  $\Rightarrow$  $\omega = 0$;  $\omega \rightarrow \infty$

 $\Rightarrow$  $\omega = 0$;  $\omega \rightarrow \infty$

*  R  $j\omega L$  $\frac{1}{j\omega C}$ $\Rightarrow \bar{Z} = R + j\omega L + \frac{1}{j\omega C} \Rightarrow \boxed{R + jX = \bar{Z}}$

(Reactive part)
reactance

resistance.

$$\Rightarrow |\bar{Z}| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$* \bar{Z} = |\bar{Z}| \angle \theta$$

$$\theta = \tan^{-1}\left(\frac{X}{R}\right); |\bar{Z}| = \sqrt{R^2 + X^2}$$

$$\Rightarrow R = |\bar{Z}| \cos \theta; X = |\bar{Z}| \sin \theta.$$

* Inverse of Impedance is called as 'Admittance'.

$$\boxed{\bar{Y} = \frac{1}{\bar{Z}}} = \frac{\bar{I}}{\bar{V}} \Rightarrow \boxed{\bar{Y} = G + jB}$$

Unit: Ω^{-1} (or) mho (or) siemen.

$G \equiv$ real part $\rightarrow$ conductance
 $B \equiv$ imaginary part $\rightarrow$ susceptance.

* Admittance

(opposite of Impedance).

In DC $\Rightarrow$ L & C played role in transient state only.

In AC circuits $\rightarrow$ they always play role.

* KCL & KVL in phasor domain:

* $V_1 + V_2 + V_3 + \dots + V_n = 0.$

$$\Rightarrow V_{m1} \cos(\omega t + \theta_1) + V_{m2} \cos(\omega t + \theta_2) + \dots + V_{mn} \cos(\omega t + \theta_n) = 0.$$

$$\text{Re} [V_{m1} e^{j\omega t} e^{j\theta_1} + V_{m2} e^{j\omega t} e^{j\theta_2} + \dots + V_{mn} e^{j\omega t} e^{j\theta_n}] = 0$$

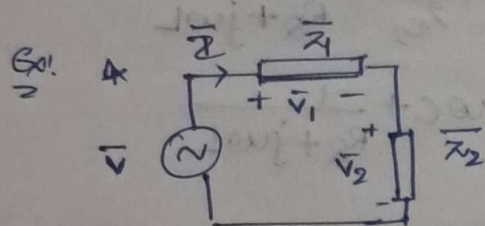
$$\Rightarrow \text{Re} [\bar{V}_1 + \bar{V}_2 + \dots + \bar{V}_n] = 0 ;$$

$$\Rightarrow \bar{V}_1 + \bar{V}_2 + \dots + \bar{V}_n = 0$$

Similarly for KVL,

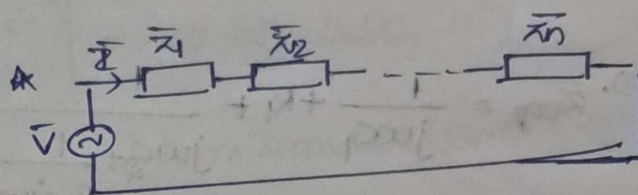
$$\bar{E}_1 + \bar{E}_2 + \dots + \bar{E}_n = 0.$$

* Impedance combinations:



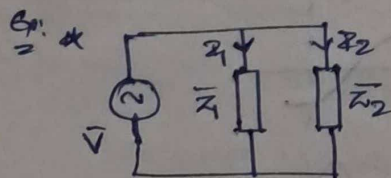
$$\bar{V} = \bar{I} (\bar{Z}_1 + \bar{Z}_2)$$

$$\bar{V}_1 = \frac{\bar{Z}_1}{\bar{Z}_1 + \bar{Z}_2} \bar{V} ; \bar{V}_2 = \frac{\bar{Z}_2}{\bar{Z}_1 + \bar{Z}_2} \bar{V}$$



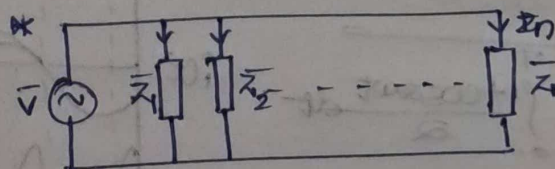
$$\bar{V} = \bar{I} (\underbrace{\bar{Z}_1 + \bar{Z}_2 + \dots + \bar{Z}_n}_{\bar{Z}}$$

* Parallel:



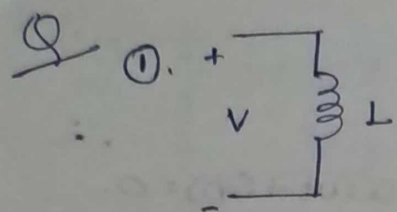
$$\bar{V} = \bar{I}_1 \bar{Z}_1 = \bar{I}_2 \bar{Z}_2$$

$$\bar{I} = \bar{I}_1 + \bar{I}_2$$



$$\frac{1}{\bar{Z}_{eq}} = \frac{1}{\bar{Z}_1} + \frac{1}{\bar{Z}_2} + \dots + \frac{1}{\bar{Z}_n}$$

$$\Rightarrow \frac{1}{\bar{Z}_{eq}} = \frac{1}{\bar{Z}_1} + \frac{1}{\bar{Z}_2} \Rightarrow \bar{I}_1 = \frac{\bar{Z}_2}{\bar{Z}_1 + \bar{Z}_2} \bar{I}$$



$$V = 12 \cos(60t + 45^\circ) = 12 \angle 45^\circ$$

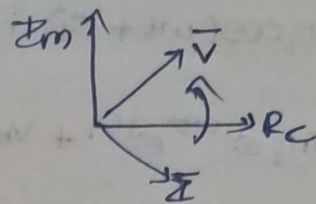
$$L = 0.1 \text{ H}$$

$$i = ?$$

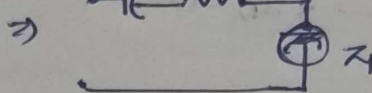
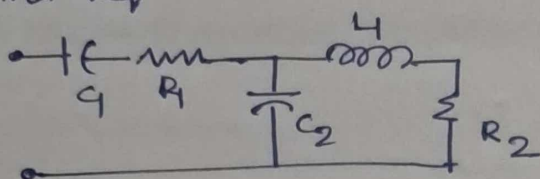
$$V = L \frac{di}{dt} \quad ; \quad \bar{X} = j\omega L = j(6)$$

$$\Rightarrow i = \frac{1}{0.1} \cdot \frac{1}{60} \sin(60t + 45^\circ) \quad (\text{A})$$

$$\Rightarrow i = 28 \sin(60t + 45^\circ) = -j \frac{(12) \angle 45^\circ}{6}$$



Q. Find $\bar{Z}_{eq}$



$$\bar{Z}_1 \Rightarrow \frac{1}{\bar{Z}_1} = \frac{1}{\bar{Z}_{C2}} + \frac{1}{R_2 + j\omega L}$$

$$\Rightarrow \bar{Z}_{eq} = \frac{1}{j\omega C_1} + R_1 + \frac{1}{j\omega C_2 + \frac{1}{R_2 + j\omega L}} \Rightarrow \frac{1}{\bar{Z}_1} = j\omega C + \frac{1}{R_2 + j\omega L}$$

POWER EXPRESSIONS:

$$P_{rms} = \sqrt{\frac{1}{T} \int_0^T i_m^2 \cos^2 \omega t dt}$$

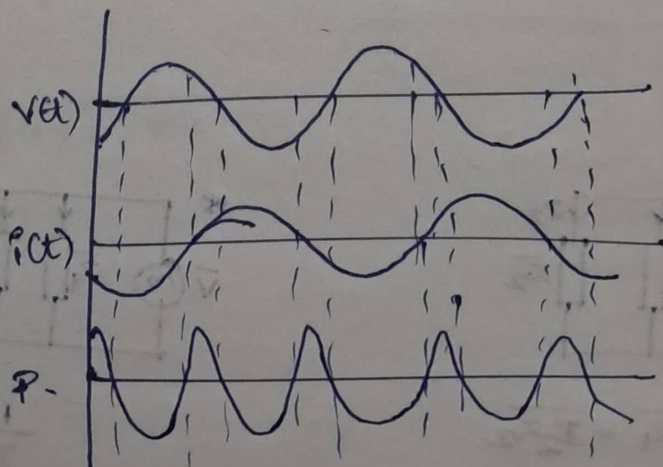
$$= \sqrt{\frac{i_m^2}{T} \int_0^T \cos^2 \omega t dt}$$

$$= \sqrt{\frac{i_m^2}{T} \int_0^T \frac{1 + \cos 2\omega t}{2} dt}$$

$$= \frac{i_m}{\sqrt{2}}$$

$$v(t) = V_m \cos(\omega t + \theta_v)$$

$$i(t) = I_m \cos(\omega t + \theta_i)$$



Instantaneous

$P(t) = v(t) i(t) \Rightarrow$ instantaneous power,

$$= V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

$$= \underbrace{\frac{V_m I_m}{2} \cos(\theta_v - \theta_i)}_{\text{constant}} + \underbrace{\frac{V_m I_m}{2} \cos(2\omega t + \theta_v + \theta_i)}_{\text{AC part with twice the frequency}}$$

constant
↓

Average power

↳ corresponds to the power absorbed by the resistors in the circuit

↳ corresponds to the power absorbed by the inductors & capacitors in the circuit.

$$P_{avg} = \frac{\int_0^T P(t) dt}{T} = \frac{1}{T} \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \frac{1}{T} \int_0^T \frac{V_m I_m}{2} \cos(2\omega t + \theta_v + \theta_i) dt$$

$$= \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) + \frac{V_m I_m}{2} (0)$$

$$\Rightarrow \boxed{P_{avg} = \frac{V_m I_m \cos(\theta_v - \theta_i)}{2}} = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i)$$

$$= V_{rms} I_{rms} \cos(\theta_v - \theta_i)$$

$$\star \frac{1}{2} \bar{V} \bar{I}^* = \frac{1}{2} V_m \angle \theta_v I_m \angle -\theta_i$$

$$= \frac{1}{2} V_m I_m = \frac{1}{2} |V| |I| e^{j(\theta_v - \theta_i)} = \frac{1}{2} V_m I_m [\cos(\theta_v - \theta_i) + j \sin(\theta_v - \theta_i)]$$

$$\Rightarrow P = \frac{1}{2} \text{Re}[\bar{V} \bar{I}^*] = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) \quad ; \quad \bar{V} = V_{rms} \angle \theta_v$$

$$\bar{I} = I_{rms} \angle \theta_i$$

Apparent Power:

$$S = V_{rms} I_{rms}$$

unit: Volt-Ampere (VA)

not watts.

* Power factor:

$$\text{Power factor} = \frac{P}{S} = \cos(\theta_v - \theta_i)$$

$$\bar{S} = \frac{\bar{V}}{\bar{I}} = \frac{V_m}{I_m} \frac{\angle \theta_v}{\angle \theta_i} = \frac{V_m}{I_m} \angle (\theta_v - \theta_i)$$

$$\Rightarrow \boxed{\bar{S} = Z_m \angle (\theta_v - \theta_i)}$$

Load Impedance angle,
⇓
Power factor angle.

$$P = V_{rms} I_{rms} \cos(\theta_v - \theta_i) \rightarrow \text{Real / Average (W)}$$

$$S = V_{rms} I_{rms} \rightarrow \text{Apparent (VA)}$$

$$\bar{S} = \bar{V} \bar{I}^* (\text{rms}) = \frac{1}{2} \bar{V} \bar{I}^* (\text{peak})$$

$$= \underbrace{V_{rms} I_{rms} \cos(\theta_v - \theta_i)}_P + j \underbrace{V_{rms} I_{rms} \sin(\theta_v - \theta_i)}_{\text{Reactive power.}}$$

(Q) (VAR)

$$\bar{V}_{rms} = \bar{Z} \bar{I}_{rms}$$

$$= (R + jX) \bar{I}_{rms}$$

$$\bar{S} = \bar{I}_{rms}^2 \bar{Z} = I_{rms}^2 (R + jX) = \underbrace{I_{rms}^2 R}_P + j \underbrace{I_{rms}^2 X}_Q$$

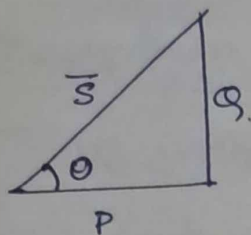
$$P = I_{rms}^2 R; \quad Q = I_{rms}^2 X$$

$$Q = 0 \rightarrow R$$

$$Q > 0 \rightarrow L$$

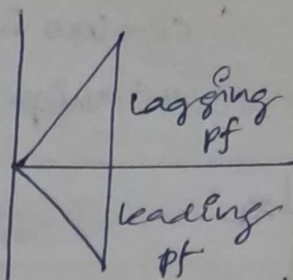
$$Q < 0 \rightarrow C$$

* Power triangle: (9/10/23)



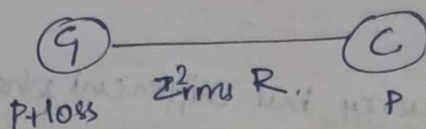
$$P = S \cos \theta$$

$$Q = S \sin \theta$$



* Power factor correction:

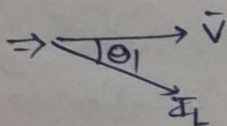
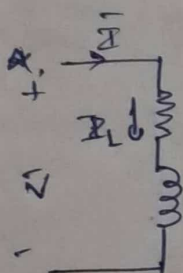
$$pf = \frac{P}{S}$$



$$P = V_{rms} I_{rms} \cos \theta \quad (\text{say } 1000W)$$

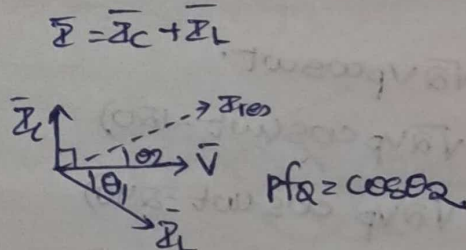
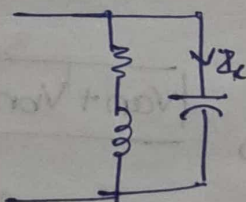
100	10	1
100	100	0.1

↳ loss is more.



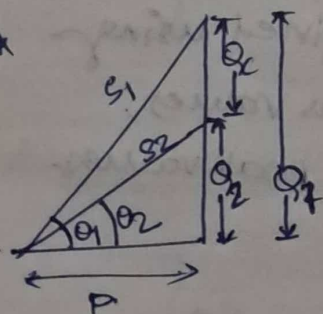
$$pf_1 = \cos \theta_1$$

→ to increase power factor, we can add cap across the load.



$$pf_2 = \cos \theta_2$$

we have to be careful about the value of capacitors.
 $\theta_2 < \theta_1$



$$pf_1 = \cos \theta_1$$

$$pf_2 = \cos \theta_2$$

$$P = S \cos \theta_1$$

$$Q_1 = S \sin \theta_1 = P \tan \theta_1$$

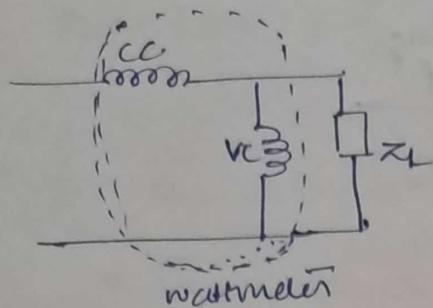
$$Q_2 = P \tan \theta_2$$

$$Q_C = Q_1 - Q_2 = P (\tan \theta_1 - \tan \theta_2)$$

$$Q_C = \frac{V_{rms}^2}{X_C} = \omega C V_{rms}^2$$

$$C = \frac{P (\tan \theta_1 - \tan \theta_2)}{\omega V_{rms}^2}$$

Power measurement:



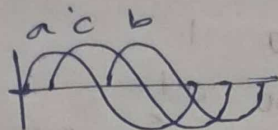
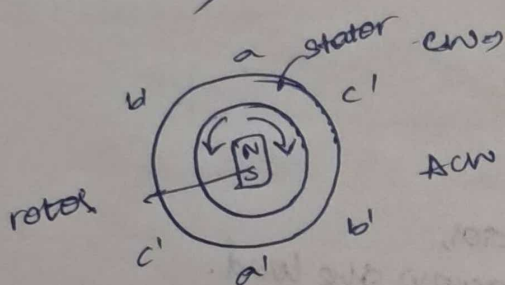
CC $\rightarrow$ low impedance.
 VC $\rightarrow$ high impedance.

POLYPHASE CIRCUITS

415V

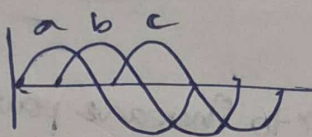
$\rightarrow$ sources are of same diff frequency but different phasers.

$\rightarrow$ transmission lines travel long distances and are of very high voltage distribution.



phasors are defined using peak values in single phase circuits.

& rms values in 3 phase circuits.



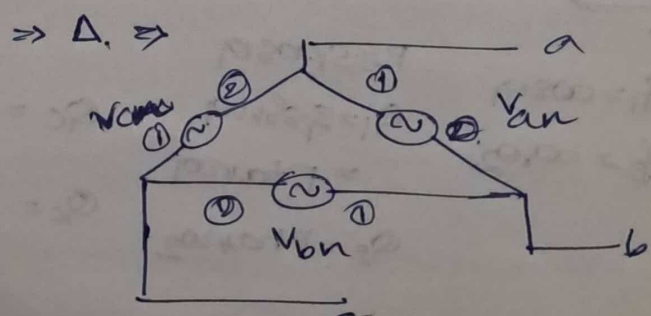
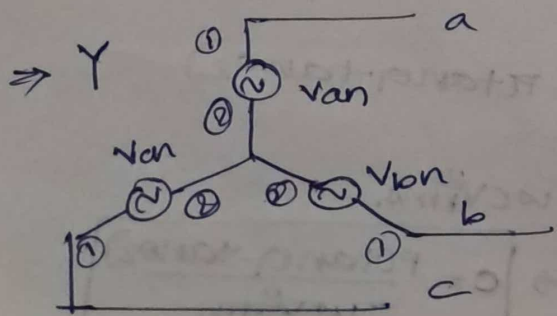
* $V_{an} = \sqrt{2} V_p \cos \omega t$
 $V_{bn} = \sqrt{2} V_p \cos (\omega t - 120^\circ)$
 $V_{cn} = \sqrt{2} V_p \cos (\omega t - 240^\circ)$
 $= \sqrt{2} V_p \cos (\omega t + 120^\circ)$

$V_{an} + V_{bn} + V_{cn} = 0$

$\bar{V}_{an} = V_p \angle 0^\circ$
 $\bar{V}_{bn} = V_p \angle -120^\circ$
 $\bar{V}_{cn} = V_p \angle 120^\circ$

defined using rms values, not peak values.

* Y & Δ connections:



* Balanced three phase voltages:

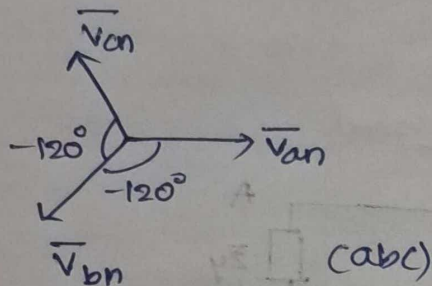
$\Rightarrow$ magnitudes of V_{an}, V_{bn}, V_{cn} are all equal. (angles shifted by 120°)

$$\Rightarrow |V_{an}| = |V_{bn}| = |V_{cn}|, \quad \Rightarrow V_{an} + V_{bn} + V_{cn} = 0.$$

(balanced ^{loads} ~~loads~~ $\Rightarrow$ angles and magnitudes all are equal).

* Phase sequences:

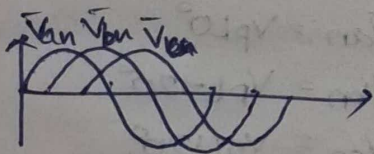
* Positive phase sequence:



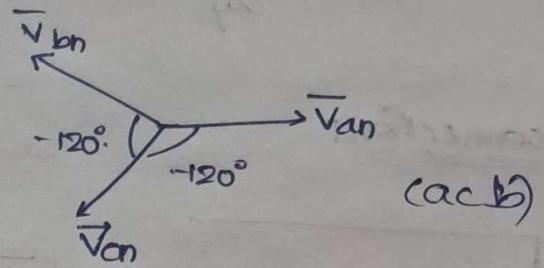
$$V_{an} = V_p \angle 0^\circ$$

$$V_{bn} = V_p \angle -120^\circ$$

$$V_{cn} = V_p \angle -240^\circ = V_p \angle 120^\circ$$



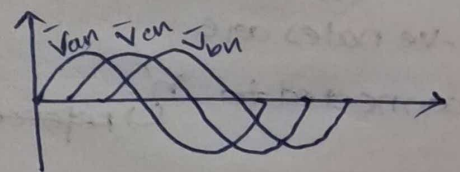
* Negative phase sequence:



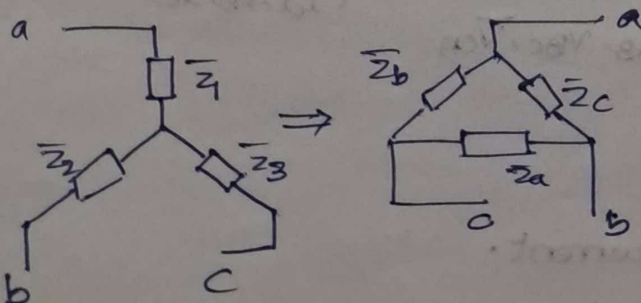
$$V_{an} = V_p \angle 0^\circ$$

$$V_{bn} = V_p \angle -240^\circ = V_p \angle 120^\circ$$

$$V_{cn} = V_p \angle -120^\circ$$



* Star $\rightarrow$ Delta conversion: ($Y \rightarrow \Delta$)



btw a & b

$$\Rightarrow \bar{Z}_1 + \bar{Z}_2 = \bar{Z}_c \parallel (\bar{Z}_a + \bar{Z}_b)$$

$$= \frac{\bar{Z}_c (\bar{Z}_a + \bar{Z}_b)}{\bar{Z}_a + \bar{Z}_b + \bar{Z}_c}$$

similarly, b & c

$$\bar{Z}_2 + \bar{Z}_3 = \frac{\bar{Z}_a (\bar{Z}_b + \bar{Z}_c)}{\bar{Z}_a + \bar{Z}_b + \bar{Z}_c}$$

by w a/c $\Rightarrow \bar{z}_1 + \bar{z}_3 = \frac{\bar{z}_b(\bar{z}_a + \bar{z}_c)}{\bar{z}_a + \bar{z}_b + \bar{z}_c}$;

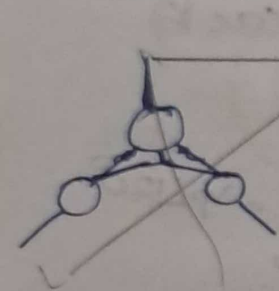
$\Rightarrow \bar{z}_1 = \frac{\bar{z}_b \bar{z}_c}{\bar{z}_a + \bar{z}_b + \bar{z}_c}$; $\bar{z}_2 = \frac{\bar{z}_a \bar{z}_c}{\bar{z}_a + \bar{z}_b + \bar{z}_c}$; $\bar{z}_3 = \frac{\bar{z}_a \bar{z}_b}{\bar{z}_a + \bar{z}_b + \bar{z}_c}$

$\bar{z}_1 = \bar{z}_2 = \bar{z}_3 = \bar{z}_Y$; $\bar{z}_a = \bar{z}_b = \bar{z}_c = \bar{z}_\Delta$

$\Rightarrow \bar{z}_Y = \frac{\bar{z}_\Delta \bar{z}_\Delta}{3\bar{z}_\Delta} = \frac{\bar{z}_\Delta}{3} \Rightarrow \boxed{\bar{z}_\Delta = 3\bar{z}_Y}$

(OR) $\bar{z}_\Delta = \frac{3\bar{z}_Y \bar{z}_Y}{\bar{z}_Y} = 3\bar{z}_Y$

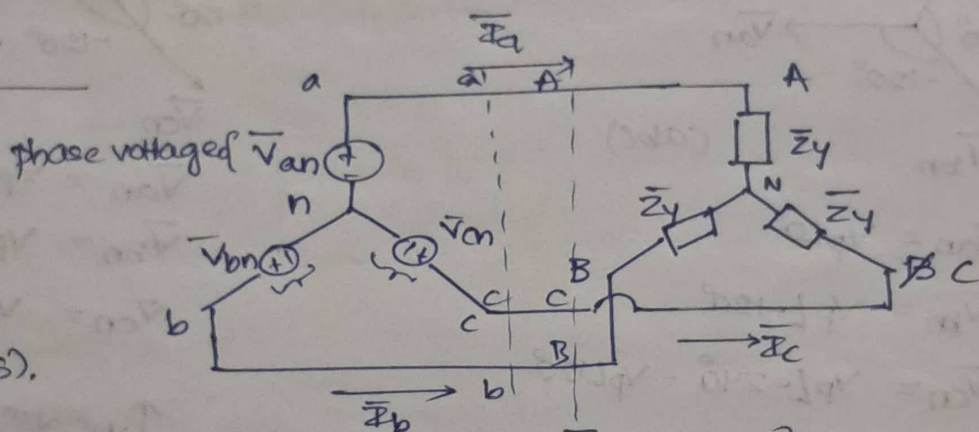
Y-Y connection:



(AC sources),

the -ve nodes are

connected to n' reference neutral.



$\bar{V}_{an} = V_p \angle 0^\circ$

$\bar{V}_{bn} = V_p \angle -120^\circ$

$\bar{V}_{cn} = V_p \angle 120^\circ$

phase voltage $\rightarrow \bar{V}_{an}, \bar{V}_{bn}, \bar{V}_{cn}, \bar{V}_{AN}, \bar{V}_{BN}, \bar{V}_{CN}$

phase current $\rightarrow$ whatever current flows through that particular component ($\bar{I}_a, \bar{I}_b, \bar{I}_c$)

line (L-L) voltage $\rightarrow \bar{V}_{ab}, \bar{V}_{bc}, \bar{V}_{ca}, \bar{V}_{AB}, \bar{V}_{BC}, \bar{V}_{CA}$

line current $\rightarrow \bar{I}_a, \bar{I}_b, \bar{I}_c$

* For Y-Y connection, line current = phase current.
line voltage $\neq$ phase voltage

$$\star \bar{V}_{ab} = \bar{V}_{an} + \bar{V}_{nb}$$

$$= \bar{V}_{an} - \bar{V}_{bn}$$

$$= V_p L \angle 0^\circ - V_p L \angle 120^\circ$$

$$= \sqrt{3} V_p L \angle 30^\circ$$

$$\Rightarrow \boxed{\bar{V}_{ab} = \sqrt{3} V_p L \angle 30^\circ}$$

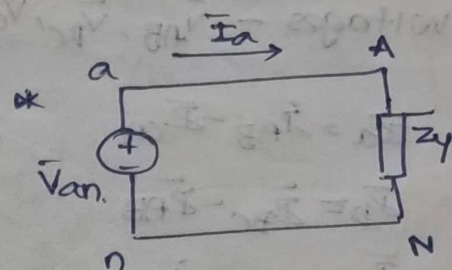
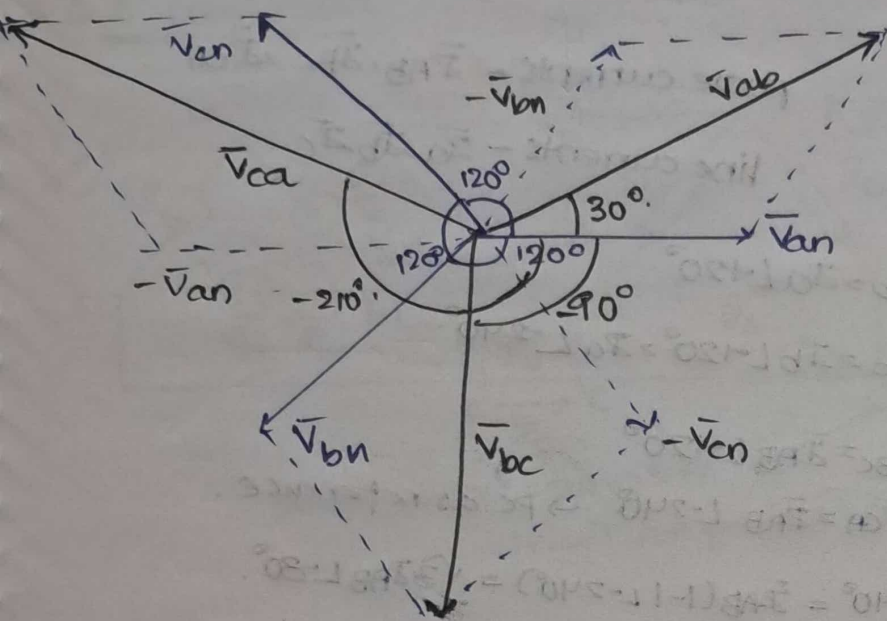
$$\star \bar{V}_{bc} = \bar{V}_{bn} - \bar{V}_{cn}$$

$$\Rightarrow \boxed{\bar{V}_{bc} = \sqrt{3} V_p L \angle 90^\circ}$$

$$(V_L = \sqrt{3} V_p)$$

$$\star \bar{V}_{ca} = \bar{V}_{cn} - \bar{V}_{an}$$

$$\Rightarrow \boxed{\bar{V}_{ca} = \sqrt{3} V_p L \angle -210^\circ}$$



$$\rightarrow \bar{I}_a = \frac{\bar{V}_{an}}{\bar{Z}_Y} = |\bar{I}_a| \angle \theta_a$$

$$\rightarrow \bar{I}_c = \bar{I}_b \angle -120^\circ = \bar{I}_a \angle -240^\circ = |\bar{I}_a| \angle -240^\circ + \theta_a$$

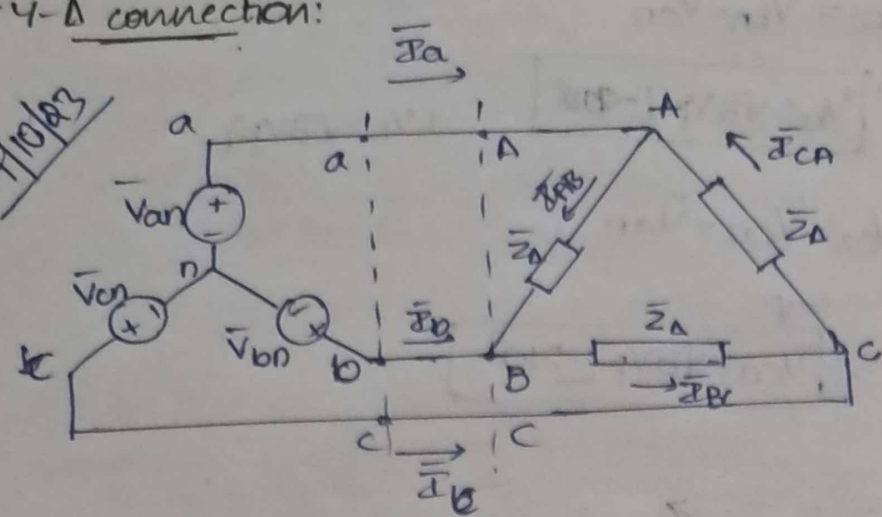
$$\rightarrow \bar{I}_b = \bar{I}_a \angle -120^\circ = |\bar{I}_a| \angle -120^\circ + \theta_a$$

$\star$ If we have unbalanced system, we have to apply KCL & KVL equations,

(PTO)

* 4-Δ connection:

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$$\begin{aligned} V_{an} &= V_p \angle 0^\circ \\ V_{bn} &= V_p \angle -120^\circ \\ V_{cn} &= V_p \angle 240^\circ \\ V_{ab}, V_{bc}, V_{ca} \\ I_a, I_b, I_c \end{aligned}$$

phase voltages - V_{AB}, V_{BC}, V_{CA}
line voltages - V_{AB}, V_{BC}, V_{CA}

phase currents - I_{AB}, I_{BC}, I_{CA}
line currents - I_a, I_b, I_c

$$\begin{aligned} \Rightarrow I_a &= I_{AB} - I_{CA} \\ I_b &= I_{BC} - I_{AB} \\ I_c &= I_{CA} - I_{BC} \end{aligned}$$

$$\begin{aligned} I_b &= I_a \angle -120^\circ \\ I_c &= I_b \angle -120^\circ = I_a \angle -240^\circ \\ I_{BC} &= I_{AB} \angle -120^\circ \\ I_{CA} &= I_{AB} \angle -240^\circ \rightarrow \text{PC as reference.} \end{aligned}$$

$$I_a = I_{AB} - I_{CA} = I_{AB} - I_{AB} \angle -240^\circ = I_{AB} (1 - \angle -240^\circ) = \sqrt{3} I_{AB} \angle -90^\circ$$

line current lags phase current by 30° .

$$\begin{aligned} \rightarrow I_b &= \sqrt{3} I_{BC} \angle -90^\circ, I_c = \sqrt{3} I_{CA} \angle -90^\circ \\ &= \sqrt{3} I_{AB} \angle -150^\circ, = \sqrt{3} I_{AB} \angle -270^\circ \end{aligned}$$

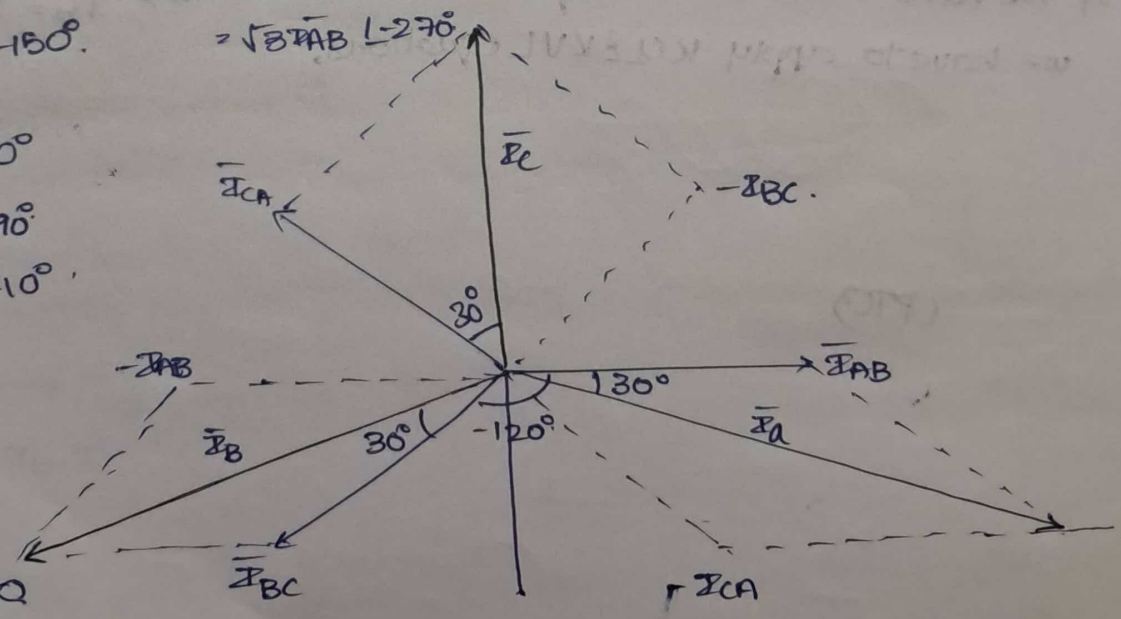
$$\begin{aligned} * V_{ab} &= \sqrt{3} V_p \angle 80^\circ \\ V_{bc} &= \sqrt{3} V_p \angle -90^\circ \\ V_{ca} &= \sqrt{3} V_p \angle -210^\circ \end{aligned}$$

$$* I_{AB} = \frac{V_{AB}}{Z_A}$$

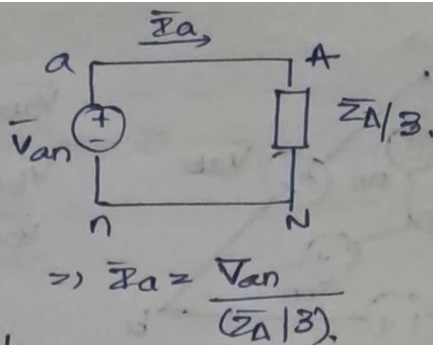
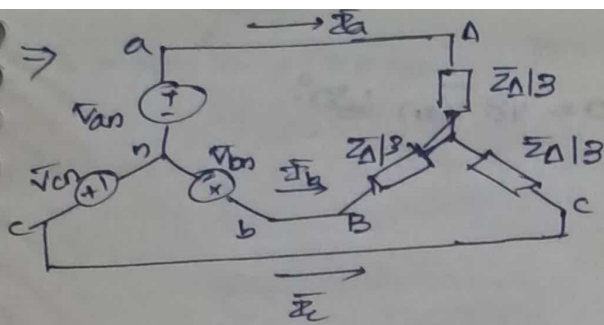
(or) KVL $\Rightarrow$

$$-V_{an} + I_{AB} Z_A + V_{bn} = 0$$

$$\Rightarrow I_{AB} = \frac{V_{an} - V_{bn}}{Z_A} = \frac{V_{AB}}{Z_A}$$



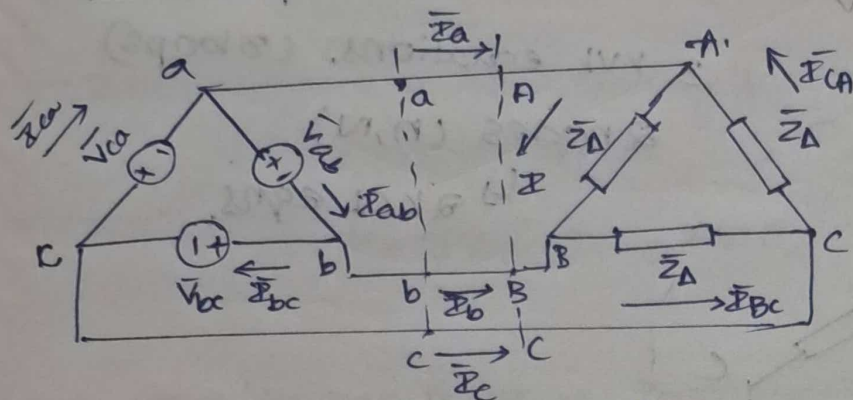
(we can also convert Δ to Y).



$$I_A = \frac{V_{an}}{(Z_A/3)}$$

don't write line current
= phase current here.

* Δ-Δ connection:

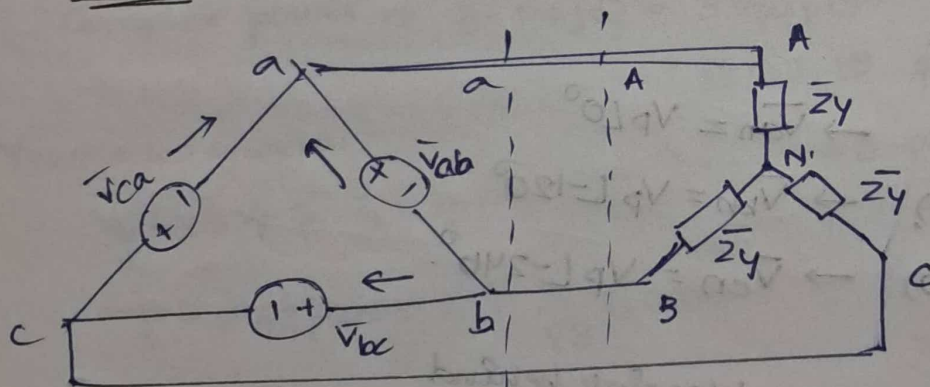


$$\begin{aligned} V_{ab} &= V_P \angle 0^\circ \\ V_{bc} &= V_P \angle -120^\circ \\ V_{ca} &= V_P \angle -240^\circ \end{aligned} \quad \left. \begin{array}{l} \text{line} \\ \text{voltage} \end{array} \right\}$$

$$I_{AB} = \frac{V_{AB}}{Z_A}$$

Single phase equivalent
circuit not possible here.

* Δ-Y connection:



$V_{ab} = V_P \angle 0^\circ$
since source
is same,
line voltages of
source remains
same

line voltage for Y:

phase voltages of load:

line currents.

$$V_{AN} = \frac{V_P}{\sqrt{3}} \angle -80^\circ$$

$$I_A = \frac{V_{AN}}{Z_Y}$$

$$V_{BN} = \frac{V_P}{\sqrt{3}} \angle -150^\circ$$

$$V_{CN} = \frac{V_P}{\sqrt{3}} \angle -270^\circ$$

$$V_{AB} = V_P \angle 0^\circ = \sqrt{3} V_{AN} \angle 30^\circ$$

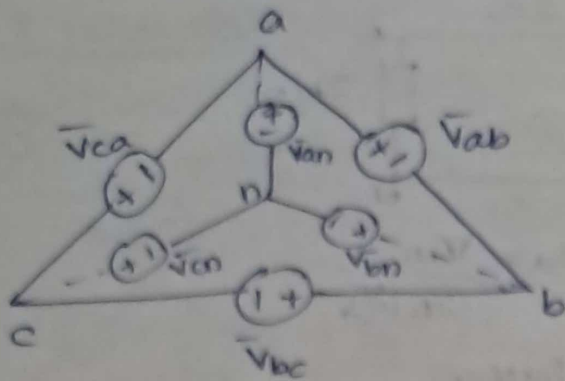
$$V_{BC} = V_P \angle -120^\circ = \sqrt{3} V_{BN} \angle 30^\circ$$

$$V_{CA} = V_P \angle -240^\circ = \sqrt{3} V_{CN} \angle 30^\circ$$

phase currents for source:

$$I_A = \sqrt{3} I_{AB} \angle -80^\circ \quad I_C = \sqrt{3} I_{CA} \angle -80^\circ$$

$$I_B = \sqrt{3} I_{BC} \angle -30^\circ$$



$$\bar{V}_{ab} = V_p L \angle 0^\circ = \sqrt{3} \bar{V}_{an} L 30^\circ$$

$$\bar{V}_{bc} =$$

$$\bar{V}_{ca} =$$

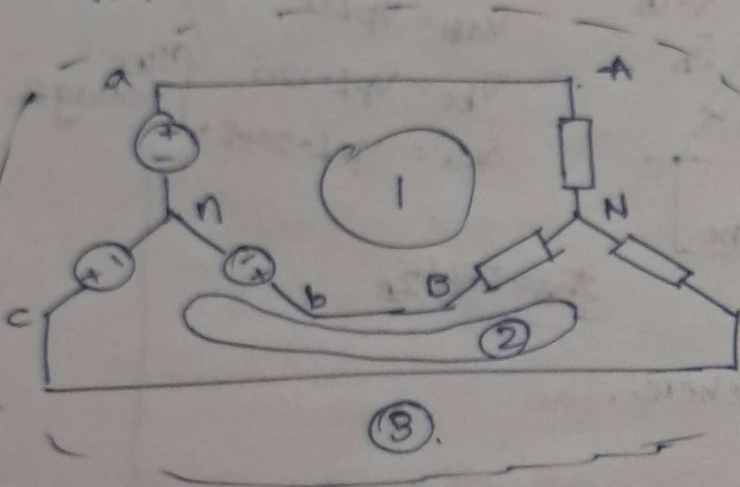
* Unbalanced three phase systems:

Apply KCL & KVL.

3 KVL equations. (3 loops)

2 nodes (n, N)

↳ 2 KCL eqns.



* Three Phase Power:

$$V_{an} = \sqrt{2} V_p \cos \omega t$$

$$\rightarrow \bar{V}_{an} = V_p L \angle 0^\circ$$

$$V_{bn} = \sqrt{2} V_p \cos(\omega t - 120^\circ)$$

$$\rightarrow \bar{V}_{bn} = V_p L \angle -120^\circ$$

$$V_{cn} = \sqrt{2} V_p \cos(\omega t - 240^\circ)$$

$$\rightarrow \bar{V}_{cn} = V_p L \angle -240^\circ$$

$$i_a = \sqrt{2} I_p \cos(\omega t - \theta) \quad \text{current lagging behind voltage.}$$

$$\bar{I}_a = I_p L \angle -\theta$$

$$i_b = \sqrt{2} I_p \cos(\omega t - 120^\circ - \theta)$$

$$\Rightarrow \bar{I}_b = I_p L \angle -\theta - 120^\circ$$

$$i_c = \sqrt{2} I_p \cos(\omega t - 240^\circ - \theta)$$

$$\bar{I}_c = I_p L \angle -\theta - 240^\circ$$

$$P_a = V_a i_a ; P_b = V_b i_b ; P_c = V_c i_c .$$

$$\Rightarrow P = P_a + P_b + P_c = 2 V_p I_p [\cos \omega t \cos(\omega t - \theta) + \cos(\omega t - 120) \cos(\omega t - \theta - 120) + \cos(\omega t - 240) \cos(\omega t - \theta - 240)]$$

$$\Rightarrow \boxed{P = 3 V_p I_p \cos \theta}$$

no time related term $\Rightarrow$ constant.

$\Rightarrow$ instantaneous power in 3 phase circuits is constant.

$\rightarrow$ instantaneous or Active / Average / Real power $\Rightarrow$ all are same for 3 phase circuits.

$$\Rightarrow P = 3 V_{rms} I_{rms} \cos \theta$$

* Active / Real / Average power $\Rightarrow P_p = 3 V_p I_p \cos \theta = \sqrt{3} V_L I_L \cos \theta$ (apply the single results with 3)

Reactive power $\Rightarrow Q_p = 3 V_p I_p \sin \theta = \sqrt{3} V_L I_L \sin \theta$

Apparent power $\Rightarrow S = 3 V_p I_p = \sqrt{3} V_L I_L$

Complex power $\Rightarrow \bar{S} = P + jQ = 3 V_p I_p \angle \theta = 3 \bar{V}_p \bar{I}_p^*$
 $= \sqrt{3} V_L I_L \angle \theta \neq \sqrt{3} \bar{V}_L \bar{I}_L^*$ (care when we take phasors - angles change)

* Y-connected system:

$$V_{ph} = \sqrt{3} V_p ; I_L = I_p$$

$$\Rightarrow P = 3 V_p I_p \cos \theta = 3 \frac{V_L}{\sqrt{3}} I_L \cos \theta$$

$$\Rightarrow \boxed{P = \sqrt{3} V_L I_L \cos \theta}$$

* Δ -connected system:

$$V_L = V_p ; I_L = \sqrt{3} I_p$$

$$P = 3 V_p I_p \cos \theta = 3 V_L \frac{I_L}{\sqrt{3}} \cos \theta$$

$$\boxed{P = \sqrt{3} V_L I_L \cos \theta}$$

don't substitute relative angle.

it is the angle b/w phase voltage & phase current

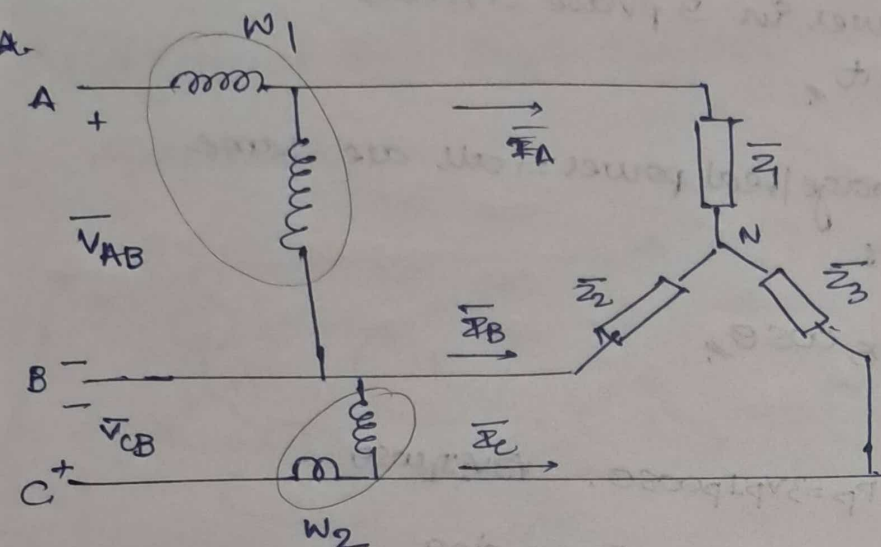
$$\begin{aligned} * \bar{S} &= 3 \bar{V}_p \bar{I}_p^* & \bar{V}_p &= \bar{I}_p \bar{Z}_p \\ &= 3 \bar{I}_p \bar{I}_p^* \bar{Z}_p & &= 3 \bar{V}_p \left(\frac{\bar{V}_p}{\bar{Z}_p} \right)^* \\ &= 3 \bar{I}_p^2 \bar{Z}_p & &= 3 \bar{V}_p^2 / \bar{Z}_p^* \end{aligned}$$

* Power measurement:

for measuring single phase $\rightarrow$ we used wattmeter.

for measuring three phase $\rightarrow$ we can use 3 wattmeters,
we can also use 2-wattmeter method.

Two wattmeter method:



$$P_1 = |\bar{V}_{AB}| |\bar{I}_A| \cos(\angle \text{angle b/w } \bar{V}_{AB} \text{ \& } \bar{I}_A)$$

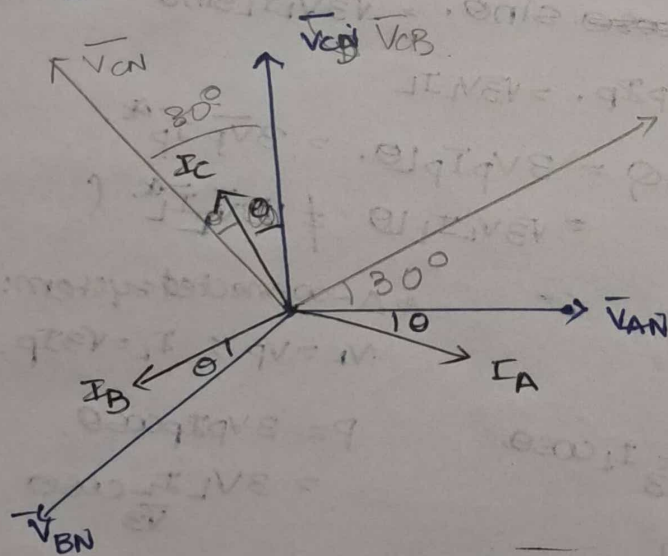
$$= V_L I_L \cos(30^\circ + \theta)$$

$$P_2 = |\bar{V}_{CB}| |\bar{I}_C| \cos(\angle \text{angle b/w } \bar{V}_{CB} \text{ \& } \bar{I}_C)$$

$$= V_L I_L \cos(30^\circ - \theta)$$

$$P = P_1 + P_2$$

$$= V_L I_L [\cos(30^\circ + \theta) + \cos(30^\circ - \theta)]$$



$$P = \sqrt{3} V_L I_L \cos \theta$$

$$P_1 - P_2 = -V_L I_L \sin \theta$$

$$= -Q/\sqrt{3}$$

$$Q = \sqrt{3}(P_2 - P_1)$$

$$S = \sqrt{P^2 + Q^2}$$

(Inductive)
* $P_2 > P_1 \Rightarrow$ lagging load

$P_2 = P_1 \Rightarrow$ resistive load

$P_2 < P_1 \Rightarrow$ leading load.
(capacitive)

$$\tan \theta = \frac{Q}{P} = \sqrt{3} \frac{P_2 - P_1}{P_2 + P_1}$$

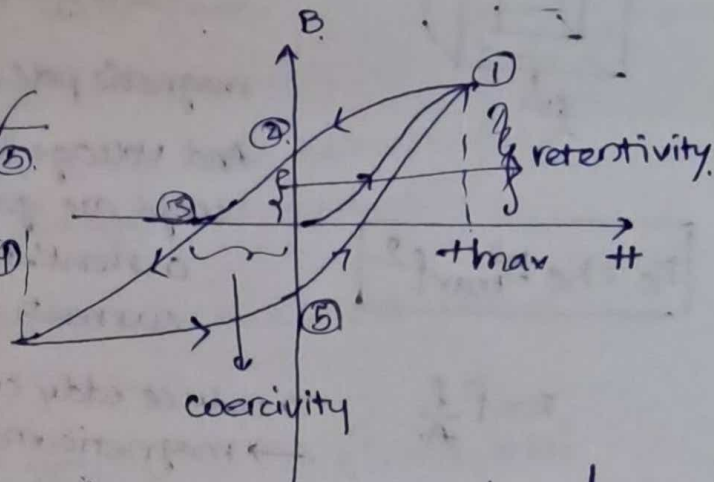
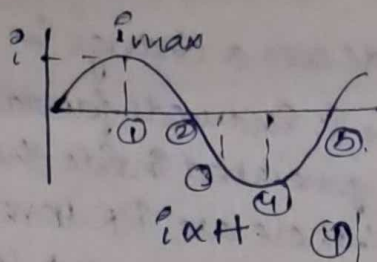
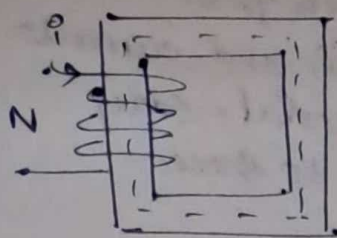
↓
power factor angle.

— QUIZ —

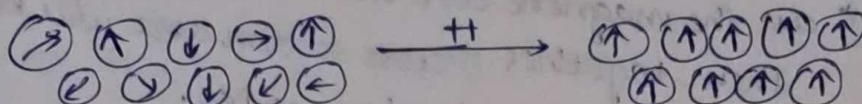
7/11/23

MAGNETIC CIRCUITS

Hysteresis Loss:



* As we apply H , the small unaligned domains start getting aligned.



→ Hysteresis Loss:

↳ proportional to how fast & frequent we trace the $B-H$ loop.

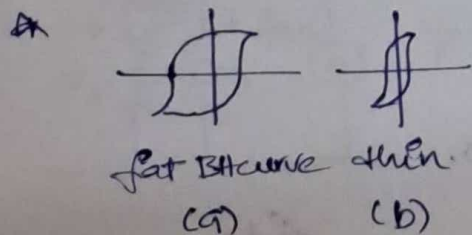
⇒ $\propto$ frequency.

* If we trace as small curve, loss will be less

↳ H & B are less.

(I_{max} is less)

Hysteresis Loss $\propto B_{max}$.



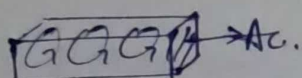
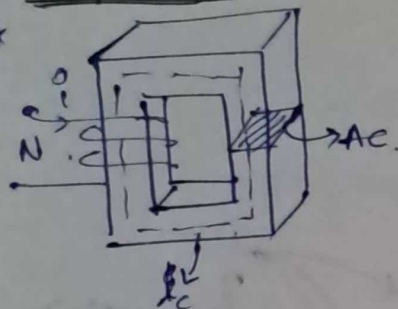
(a) will have more losses compared to (b).

$$P_H = K_h B_{max}^m f$$

$K_h, m \Rightarrow$ dependent on material.

* Hysteresis loss is minimised for magnetic materials.

Eddy currents:



↳ acting like a conductor.

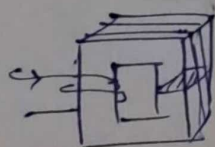
magnetic pole sees a change in magnetic field
And voltage is induced (Faraday's law) and current loops are generated inside the material $\Rightarrow$ eddy currents. There are $i^2 R$ losses due to these currents $\Rightarrow$ Eddy current losses.

$$P_e = K_e B_{max}^2 f^2$$

$$R = \rho \frac{l}{A}$$

To reduce eddy current losses $\rightarrow$

$\rightarrow$ magnetic material should have high resistivity
 $\rightarrow$ induced currents will reduce.



$\rightarrow$ cut the magnetic core into thinner sheets & stack them together $\Rightarrow$ cross section area will be smaller for each sheet $\Rightarrow$ resistance $\uparrow$.

$\rightarrow$ The sheets are called 'laminates' or 'laminations'.

↳ 50Hz transformers use these widely.
material: CRGO.

* High frequency $\Rightarrow$ ferromagnetic materials are used.

(Lamination is not used 'cause they are too brittle for it)

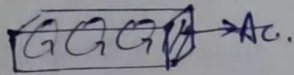
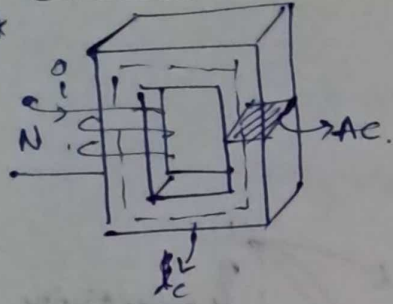
$$P_c = P_H + P_e$$

* Magnetically coupled circuits:

Ex: Transformers, wireless charging of phones.

$\rightarrow$ no electrical connection b/w the primary & secondary sides.

Eddy currents:



↳ acting like a conductor.

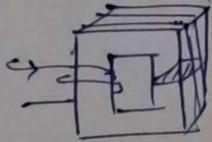
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*

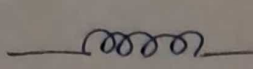
$$P_c = P_H + P_e$$

* Magnetically coupled circuits:

Ex: Transformers, wireless charging of phones.

$\rightarrow$ no electrical connection b/w the primary & secondary sides

* Inductance:

 $V = L \frac{di}{dt}$

A current carrying conductor will produce some flux.

Flux linkage $\lambda = N\phi$

$$\rightarrow L = \frac{\lambda}{i} = \frac{N\phi}{i} = \frac{NBA}{i}$$

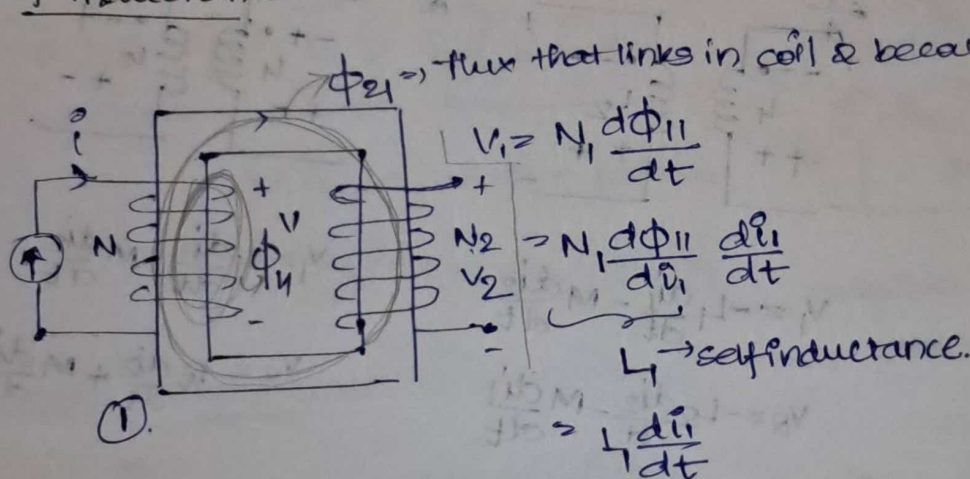
'L'
 $\Rightarrow$ It is the ratio between flux linkage and current.

$\Rightarrow$ whenever there is flux, there is inductance associated with it.

$$\star L = \frac{NBA}{i} = \frac{N\mu H A}{i} = N\mu A \frac{N}{l_c} = \frac{N^2}{(l_c/\mu A)} \quad (\because H = N^2/l_c)$$

$$\Rightarrow R = l_c/\mu A. \text{ (Reluctance).}$$

* Self-inductance:

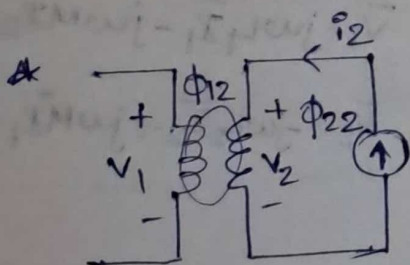


$$V_2 = N_2 \frac{d\phi_{21}}{dt}$$

$$= N_2 \frac{d\phi_{21}}{d i_1} \frac{d i_1}{dt}$$

$M \rightarrow$ Mutual inductance

$$= M_{21} \frac{d i_1}{dt}$$



$$V_2 = N_2 \frac{d\phi_{22}}{dt}$$

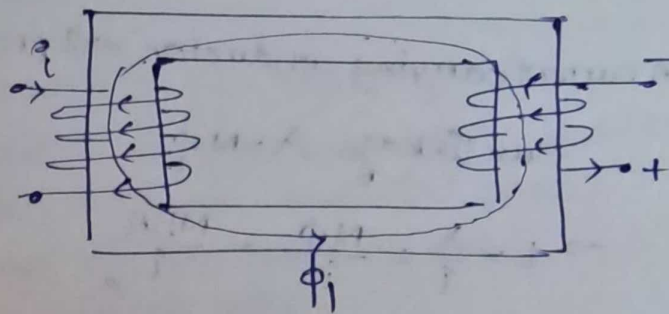
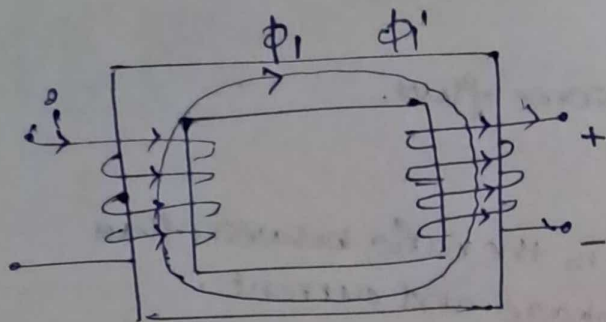
$$= L_2 \frac{d i_2}{dt}$$

$$V_1 = N_1 \frac{d\phi_{12}}{dt}$$

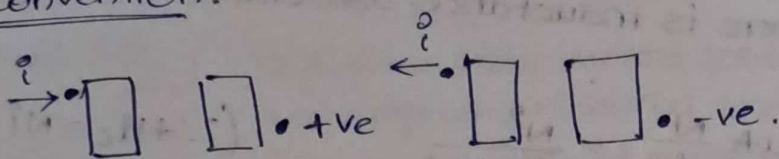
$$= M_{12} \frac{d i_2}{dt}$$

$\Rightarrow \boxed{M_{21} = M_{12} = M}$ (proof will be discussed in next class)

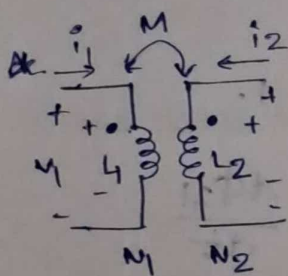
Polarity of induced voltage:



Dot convention:



If a current enters the dot terminal of 1st coil, the polarity at the dot terminal of 2nd coil is positive. ~~& vice versa.~~

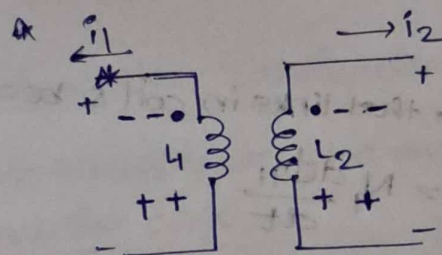


$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$v_2 = L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

$$\bar{v}_1 = j\omega L_1 \bar{i}_1 + j\omega M \bar{i}_2$$

$$\bar{v}_2 = j\omega L_2 \bar{i}_2 + j\omega M \bar{i}_1$$

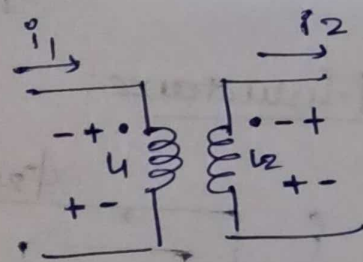


$$v_1 = -L_1 \frac{di_1}{dt} - M \frac{di_2}{dt}$$

$$v_2 = -L_2 \frac{di_2}{dt} - M \frac{di_1}{dt}$$

$$\bar{v}_1 = -j\omega L_1 \bar{i}_1 - j\omega M \bar{i}_2$$

$$\bar{v}_2 = -j\omega L_2 \bar{i}_2 - j\omega M \bar{i}_1$$

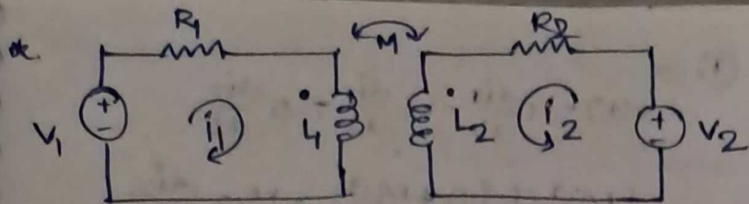


$$v_1 = L_1 \frac{di_1}{dt} - M \frac{di_2}{dt}$$

$$v_2 = -L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

$$\bar{v}_1 = j\omega L_1 \bar{i}_1 - j\omega M \bar{i}_2$$

$$\bar{v}_2 = -j\omega L_2 \bar{i}_2 + j\omega M \bar{i}_1$$

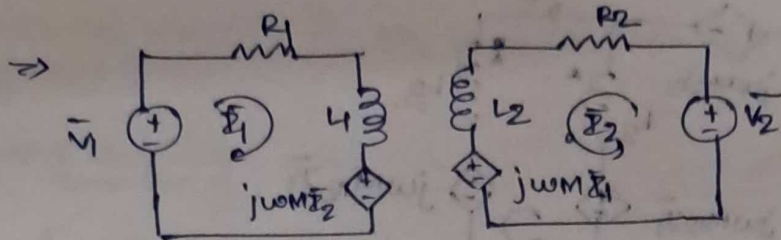


$$V_1 = i_1 R_1 + L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

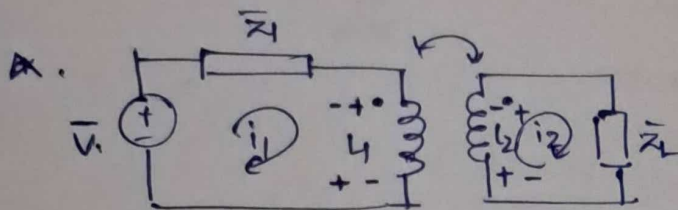
$$V_2 = i_2 R_2 + L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

$$\Rightarrow \bar{V}_1 = R_1 \bar{I}_1 + j\omega L_1 \bar{I}_1 + j\omega M \bar{I}_2$$

$$\bar{V}_2 = R_2 \bar{I}_2 + j\omega L_2 \bar{I}_2 + j\omega M \bar{I}_1$$

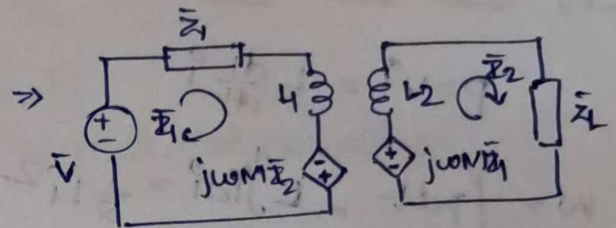


(equivalent circuit of representing a mutual inductance drop with independent voltage source drops).



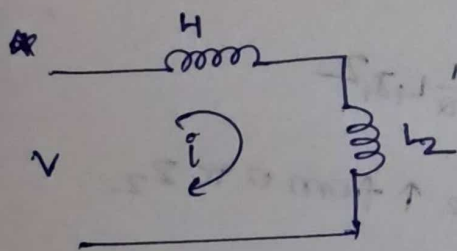
$$\bar{V} = \bar{Z}_1 \bar{I}_1 + j\omega L_1 \bar{I}_1 - j\omega M \bar{I}_2$$

$$0 = \bar{Z}_L \bar{I}_2 + j\omega L_2 \bar{I}_2 - j\omega M \bar{I}_1$$



(whatever polarity on dot due to other coil comes first)

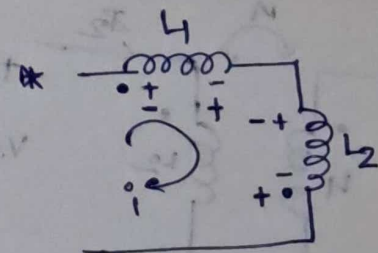
* Equivalent Inductance:



(no coupling)

$$V = (L_1 + L_2) \frac{di}{dt}$$

$$\Rightarrow \boxed{L_{eq} = L_1 + L_2}$$

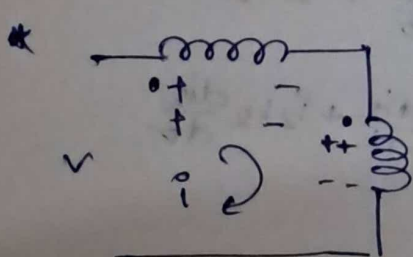


(coupling)

$$V = L_1 \frac{di}{dt} + L_2 \frac{di}{dt} - M \frac{di}{dt} - M \frac{di}{dt}$$

$$\Rightarrow V = (L_1 + L_2 - 2M) \frac{di}{dt}$$

$$\Rightarrow \boxed{L_{eq} = L_1 + L_2 - 2M}$$

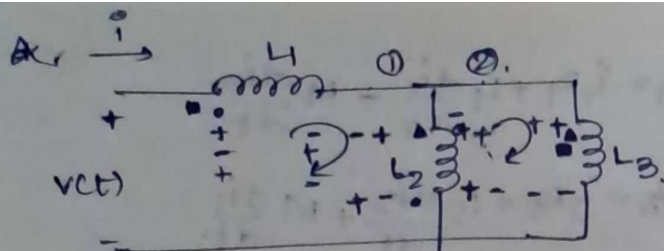


(coupling)

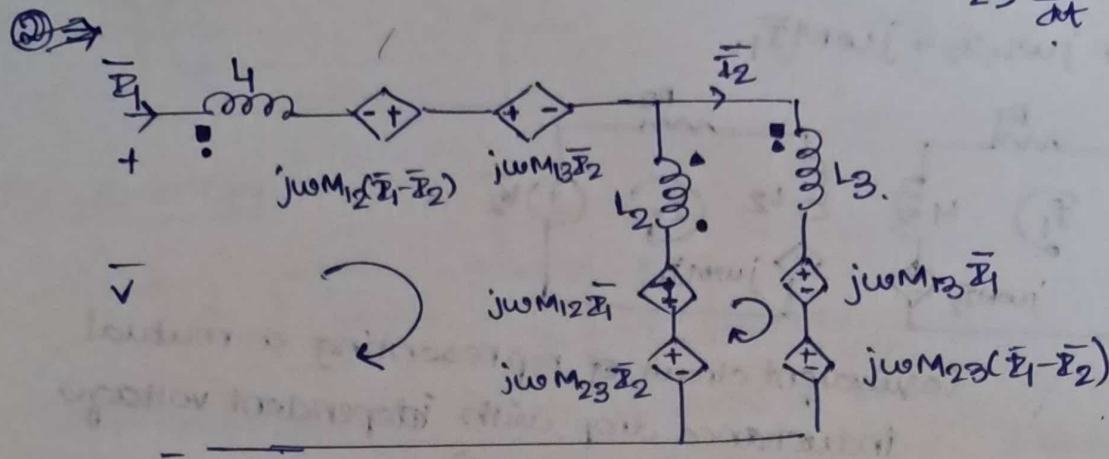
$$V = (L_1 + L_2) \frac{di}{dt} + M \frac{di}{dt} \times 2$$

$$= (L_1 + L_2 + 2M) \frac{di}{dt}$$

$$\boxed{L_{eq} = L_1 + L_2 + 2M}$$



$$\textcircled{1} \Rightarrow v(t) - L_1 \frac{di_1}{dt} - L_2 \frac{di_1}{dt} + L_3 \frac{di_2}{dt} + M_{12} \frac{d}{dt}(i_1 - i_2) + M_{12} \frac{di_1}{dt} - M_{13} \frac{di_2}{dt} - M_{23} \frac{di_2}{dt} = 0$$

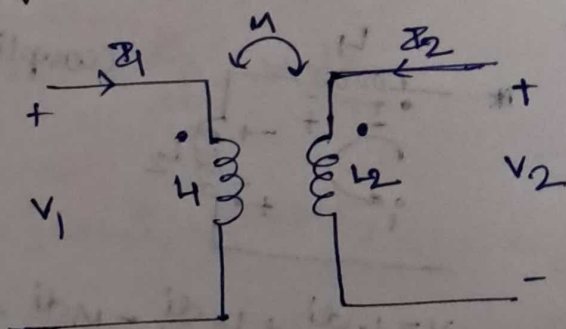


Energy stored in Inductor:

$$P = v i = L \frac{di}{dt} i$$

$$W = \int_{-\infty}^t P dt = L \int_{-\infty}^t i di = \frac{1}{2} L i^2$$

Energy stored in a coupled circuit:



Step 1: $i_2 = 0$; i_1 ↑ from 0 to I_1
 $P_1 = v_1 i_1 = L_1 \frac{di_1}{dt} i_1$

$$v_2 i_2 = 0$$

$$W_1 = \int P_1 dt = \frac{1}{2} L_1 I_1^2$$

Step 2: $i_1 = I_1$; i_2 ↑ from 0 to I_2

$$P_2(t) = i_1 M_{12} \frac{di_2}{dt} + i_2 v_2$$

$$\Rightarrow W_2 = M_{12} I_1 I_2 + \frac{1}{2} L_2 I_2^2$$

$$= I_1 M_{12} \frac{di_2}{dt} + i_2 L_2 \frac{di_2}{dt}$$

$$\Rightarrow W = W_1 + W_2 = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M_{12} I_1 I_2$$

→ If current dirn is reversed in one of the inductors

(or if dot positions are different)

$$W = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 - M i_1 i_2$$

$$\Rightarrow \boxed{W = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 \pm M i_1 i_2}$$

(say step 1 & 2 are reversed $\Rightarrow M_{12} = M_{21} \Rightarrow M_{12} = M_{21} = M$)

* Coefficient of coupling:

$$\frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 - M i_1 i_2 \geq 0$$

$$\frac{1}{2} (i_1 \sqrt{L_1} - i_2 \sqrt{L_2})^2 + i_1 i_2 (\sqrt{L_1 L_2} - M) \geq 0$$

$\Downarrow$
+ve

+ve

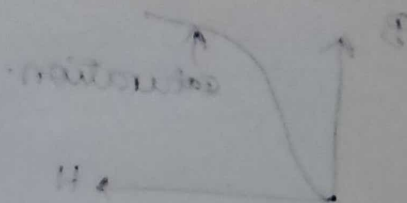
$$\Rightarrow M \leq \sqrt{L_1 L_2}$$

$$\Rightarrow \boxed{M = K \sqrt{L_1 L_2}}, \quad \boxed{0 \leq K \leq 1}$$

coefficient of coupling.

$K=1 \Rightarrow$ tightly coupling.

$K=0 \Rightarrow$ loose coupling.



$$\frac{dW}{dt} = \dots$$

